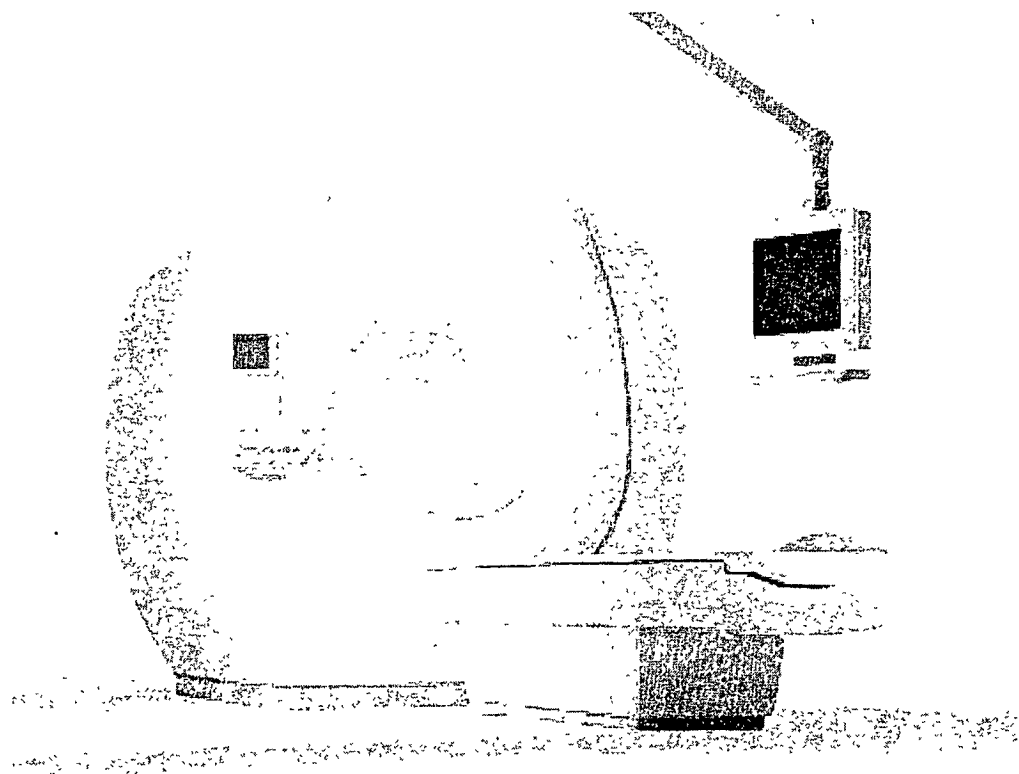


Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

Section 22

MRL

MEASUREMENT RESULTS LIST**GYROSCAN INTERA SYSTEMS**

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

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Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

Intera 0.5 T	Intera 1.0 T		Intera 1.5 T		
Standard	Omni	Power	Omni	Power	Master

MEASUREMENT RESULTS LIST

HOSPITAL _____

ADDRESS _____

COUNTRY _____

CHECKED BY _____

SYSTEM NUMBER : _____

INSTALLATION START : _____

FINISH : _____

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

1. CONFIGURATION

BASIC SYSTEM

Computer		
MR/VMS	R	date
ASW	R	date
TSW	R	date:
Reconstructor	Serial number	:
BDAS	Serial number	:
RF Power Amplifier	Type	:
	Serial number	:
Body coil	Serial number	:
Head coil	Serial number	:
Gradient Amplifier	Type	:
	Serial number	:
Only for Master gradient slave	Serial number	:
Gradient coil	Type	:
	Serial number	:
Magnet (IGC/AISA)	Serial number	:

OPTIONS:

Surface coils:	R1	yes/no
	C4	yes/no
	E1	yes/no
	BWA	yes/no
	TMJ	yes/no
	Quad_Knee	yes/no
	Quad_Neck	yes/no
	TL Spine	yes/no
	Breast	yes/no
	Endo	yes/no
	Synergy option	yes/no
	Synergy spine coil	yes/no
	Synergy cardiac coil	yes/no
	Synergy body coil	yes/no
	Synergy flex M coil	yes/no
	Synergy pediatric	yes/no
Easy Vision MR/CT		yes/no
Networking		yes/no
Hardcopy unit		yes/no
	Type:	

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

3. MAINS DISTRIBUTION CHECK LIST FOR POWER AND MASTER

Insulation measurements.

Resistance between:	Spec.	Measured
Magnet - RF enclosure	>200 k Ω	
Patient Support - RF enclosure	>200 k Ω	

Earth bounding measurement (MDU).

Clip between:		Spec.	Measured
Central PE	enclosure of		
MDU Extended	Gradient supply	<100 m Ω	
MDU Extended	RF amplifier	<100 m Ω	
MDU Extended	NTDAC	<100 m Ω	
MDU Extended	RMMU	<100 m Ω	
MDU Extended	Patient ventilation unit	<100 m Ω	
MDU Extended	Console equipment cab.	<100 m Ω	
MDU Extended	System filter box	<100 m Ω	
MDU Extended	Head exchanger	<100 m Ω	

Earth bounding measurement Monitor console.

Clip between:		Spec.	Measured
Central PE	Central PE		
Monitor console	Console trolley	< 20 m Ω	
Monitor console	BE Misc. box	< 20 m Ω	

Earth bounding measurement (System filter box).

Clip between:		Spec.	Measured
Central PE	Central PE		
System filter box	NTDAC	< 50 m Ω	
System filter box	magnet	< 20 m Ω	
System filter box	patient table base	< 100 m Ω	
System filter box	Physiology unit in patient table box	< 20 m Ω	
System filter box	ring frame front	< 20 m Ω	
System filter box	ring frame rear	< 20 m Ω	
System filter box	PFEI	< 20 m Ω	
System filter box	PICU front	< 20 m Ω	
System filter box	Cold head	< 20 m Ω	

Leakage current measurements.

System mode	Spec.	Measured
Standby	< 10 mA	
PIQT	< 15 mA	

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T		
Standard	Omni	Power	Omni	Power	Master	

4. MAGNET RESULTS

4.1. GENERAL MAGNET INSTALLATION ASPECTS.

Helium usage during installation.

	LHe level at arrival	LHE level end install.
date		
level		
Total amount of LHe used:		

ERDU (battery/connection/quench): OK : yes/no

Final settings magnet power supply and F0:

Current potmeter: _____
 Current on display. _____
 Field measured with tesla meter: _____
 F0: _____

Quench record.

Date:	LHe level		Action:
	Before	After	

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

4.2. SHIM RESULTS:

Insert a print-out of all the shim plots in this MRL (enclosure 1).

4.3. EDDY CURRENT CHECK

Insert a printout of the eddy current adjustment in this MRL (enclosure 2).

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T		
Standard	Omni	Power	Omni	Power	Master	

5. SAFETY CHECKS

Safety checks results.

Check	correct
Pick-up coil switches off	yes / no
Scan stops, when "STOP SCAN" on the operators console is pressed	yes / no
Scan stops, when "STOP SCAN" on the front operator panel is pressed	yes / no
Scan stops, when "STOP SCAN" on the rear operator panel is pressed	yes / no
Scan stops, when RF-door is opened	yes / no
Finger protection test	yes / no
Manual decoupling device	yes / no
Manual function	yes / no
Cryo cooler switch off function	yes / no
Helium overpressure	yes / no
Airflow circuit	yes / no
Gradient coil temperature sensors detection	yes / no
Patients alarm tests	yes / no

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

6. COIL PARAMETER MEASUREMENTS RESULTS

6.1. BODY COIL PARAMETERS 0.5T

Parameters	Measured	Specification	Q-factor	Specification
Rod 3 - 11 (F1)		$< \pm 10$ kHz		> 400
Rod 7 - 15 (F2)		$< \pm 10$ kHz $F2 = F1 \pm 2.5$ kHz		> 400
Actual bodycoil freq.		n.a.		n.a.

6.2. HEAD COIL PARAMETERS 0.5T

Probe position		Parameters with specification		Measured values
Step 1) L8 - L12	Jumper position: W1: Removed W2: Position 2	Q(1)	> 350	
		Voltage U ₁		
		F(1)	< ± 10 kHz	
Step 2) L5 - L11	Jumper position: W1: Position 2 W2: Removed	Q(2)	> 350	
		Voltage U ₂		
		F(2)	< ± 10 kHz	
Step 3) L8 - L12	Jumper position: W1: Position 2 W2: Removed	Voltage U ₃		
		Att(1)	typical value > 15 dB	
Step 4) L5 - L11	Jumper position: W1: Removed W2: Position 2	Voltage U ₄		
		Att(2)	typical value > 15 dB	
Both jumpers positioned		F(head)	n.a.	
		Q(head)	n.a.	

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

6.3. BODY COIL PARAMETERS 1.0T OMNI

Determination of the bodycoil parameters:

Parameters	Freq. offset	Specification	Q-factor	Specification
Actual bodycoil freq.		n.a.		$Q \geq 300$

6.4. BODY COIL PARAMETERS 1.0T POWER

Determination of the bodycoil parameters:

Parameters	Freq. offset	Specification	Q-factor	Specification
Actual bodycoil freq.		n.a.		$Q \geq 250$

6.5. HEAD COIL PARAMETERS 1.0T

Probe position	Parameters with specification		Measured values
Step 1) L8 - L12	Jumper position: W1: Position 2 W2: Removed	Q(1)	> 250
		Voltage U_1	
		F(1)	< ± 25 kHz
Step 2) L5 - L11	Jumper position: W1: Removed W2: Position 2	Q(2)	> 250
		Voltage U_2	
		F(2)	< ± 25 kHz
Step 3) L8 - L12	Jumper position: W1: Removed W2: Position 2	Voltage U_3	
		Att(1)	typical value > 17 dB
Step 4) L5 - L11	Jumper position: W1: Position 2 W2: Removed	Voltage U_4	
		Att(2)	typical value > 17 dB
Both jumpers positioned (reference values)		F(head)	n.a.
		Q(head)	n.a.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T		
Standard	Omni	Power	Omni	Power	Master	

6.6. BODY COIL PARAMETERS 1.5T OMNI

Determination of the bodycoil parameters:

Parameters	Freq. offset	Specification	Q-factor	Specification
Actual bodycoil freq.		n.a.		$Q \geq 190$

6.7. BODY COIL PARAMETERS 1.5T POWER AND MASTER

Determination of the bodycoil parameters:

Parameters	Freq. offset	Specification	Q-factor	Specification
Actual bodycoil freq.		n.a.		$Q \geq 170$

6.8. HEAD COIL ACS-NT EL_LOAD PARAMETERS

		Parameters with specification		Measured values
Step 1) L8 - L12	Jumper position: W1: Removed W2: Position 2	Q(1)	Typical value > 90	
		Voltage U ₁		
		F(1)	< ± 25 kHz	
Step 2) L5 - L11	Jumper position: W1: Position 2 W2: Removed	Q(2)	Typical value > 90	
		Voltage U ₂		
		F(2)	< ± 25 kHz F(1)-F(2) ≤ 25kHz	
Step 3) L8 - L12	Jumper position: W1: Position 2 W2: Removed	Voltage U ₃		
		Att(1)	typical value > 15 dB	
Step 4) L5 - L11	Jumper position: W1: Removed W2: Position 2	Voltage U ₄		
		Att(2)	typical value > 15 dB	
Both jumpers positioned (reference values)		F(head)	n.a.	
		Q(head)	n.a.	

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

7. IMAGE QUALITY SPECIFICATIONS

The image quality will be specified for:

- Pre_performance test
- Bottom_line specifications for transversal head-coil scans (HEADT)
- Bottom_line specifications for transversal body-coil scans (BODYT).
- Surface coil scans.
- Periodic Image Quality Test scans

The image quality spec's are distributed via "spec-files". From release 4.5 onwards, two spec-files are used. One for the Standard (ST) coils and one for the New Technology (NT) coils. For release R6.1.2 and the derived maintenance releases R5.6.2 and R4.6.2 the initial I.Q. Spec files are installed during the installation of the ASW software. Updates of these spec-files can be found on the BBS in the "download files" area of the product group MR and on the DSS-II CD-ROM subdirectory D:\VARIOUS.MRI. The spec-files must be loaded into the GYRO\$SPT_SPEC directory and renamed using the following naming convention:

Spec_file: aaRbVcLd_ee_Lff_gg.SPEC

aa : T5, T10, or ACS
b : Release nr.
c : Version nr.
d : Level nr.
ee : PowerTrak configuration (00 = PT1000, 01 = PT1000 Advanced, 02 = PT3000, 03 = PT6000).
ff : Level spec-file.
gg : ST or NT coils

Example: the file name for R6.1.2 PowerTrak 6000, NT Coils, first update = ACSR6V1L2_03_L01_NT.SPEC

Please refer to chapter "Operator console" paragraph "connecting a service PC" to the host and "copying files from/to the NT system to/from the service PC" for more information about how to transfer the files.

Also from release 4.5 onwards, the integrity of the spec-files is ensured by a Cyclic Redundancy Check (CRC). A warning is given when the file is used -after unauthorised editing- to produce a MRL (measurement result list). MRL values which are out of spec are marked with an asterisk and values which are not specified with "na" in the output file (MRL with specifications).

Off-centre

The specifications are only valid if the offcentre distance in the cc-direction is $\leq 5\text{mm}$ for BODYT; this will be automatically detected by the scout scans of SPT.

FloodField Uniformity:

When the radius of the 10% area is not mentioned, this means that this parameter is not specified.

Slice profile:

The specifications for SLP are only valid when the correct image type is used:

MXV for SE and IR

Modulus for FFE

The phantom type should be MM1 (1mm ramp) while the used slice thickness for all the headscans is always at least 5mm.

The position shift should always be within 3mm

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T		
Standard	Omni	Power	Omni	Power	Master	

7.1. PRE-PERFORMANCE TEST

7.1.1. ARTEFACT LEVEL.

The values for the artefact level as given in the tables below are NOT analyzed automatically by the software but has to be done manually as described in the Acceptance Section, and therefore are not in the spec_file.

Artefact level with IQT:

Protocol		H01	H02
Artefact level			
	spec	echo 1	
total (%)	<3	___	___
	spec	echo 2	
total (%)	<3	___	___
	spec	echo 3	
total (%)	<3	___	___
	spec	echo 4	
total (%)	<3	___	___

Protocol.		B01	B02	B03
Artefact level				
	spec	echo 1		
total (%)	<3	___	___	___
	spec	echo 2		
total (%)	<3	___	___	___
	spec	echo 3		
total (%)	<3	___	___	___
	spec	echo 4		
total (%)	<3	___	___	___

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

7.1.2. CONNECTOR TEST.

Protocol: H11, slice 3, echo 1	
Connector	SNR(B)
1	_____
2	_____
Calculate the deviation according to the formula below	_____ < 0.05

$$deviation[n] = \frac{|SNR(B)[n] - average|}{average} \text{ and } average = \frac{\sum_{n=1}^m SNR(B)[n]}{m}$$

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

7.2. IMAGE QUALITY MEASUREMENTS.

Included a print-out of the MRL as enclosure 3.

8. LIST OF ENCLOSURES.

Make sure that the following enclosures are included with this MRL.

Enclosure 1: results shimming

Enclosure 2: results eddy current adjustment

Enclosure 3: print-out of Image Quality MRL.

Enclosure 4: Tuned hardware parameters